BC 3
Taylor Series Review

Name: key

1. Use Maclaurin series to write power series expanded about $x_0 = 0$ (in simplified Σ-notation) for each of the following functions.

a. $f(x) = \dfrac{3x}{2-5x}$

$= \dfrac{3x}{2\left(1-\frac{5}{2}x\right)} = \dfrac{\frac{3}{2}x}{1-\frac{5}{2}x}$ Infinite Geo Series $a_1 = \frac{3}{2}x$, $r = \frac{5}{2}x$

$= \sum_{k=0}^{\infty} \left(\frac{3}{2}x\right)\left(\frac{5}{2}x\right)^k = \sum_{k=0}^{\infty} \left(\frac{3}{2}\right)\left(\frac{5}{2}\right)^k x^{k+1}$

b. $f(x) = \dfrac{x^3}{e^{2x-1}}$ (Be careful with this one!)

$= \dfrac{x^3}{e^{2x}e^{-1}} = \dfrac{x^3 e}{e^{2x}}$ ← constant $= (ex^3)e^{-2x} = ex^3\left(1+(-2x)+\frac{(-2x)^2}{2!}+\frac{(-2x)^3}{3!}+\ldots\right)$

Not in the format of Maclaurin series yet. →

$= ex^3 \sum_{k=0}^{\infty} \dfrac{(-2x)^k}{k!} = \sum_{k=0}^{\infty} (ex^3)\dfrac{(-2x)^k}{k!} = \boxed{\sum_{k=0}^{\infty} \dfrac{(-1)^k e \cdot 2^k x^{k+3}}{k!}}$

2. Write the first four nonzero terms of the Taylor series for $f(x) = \sin(x)$, expanded about $x_0 = \dfrac{3\pi}{4}$.

$f\left(\frac{3\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$f'(x) = \cos(x)$ $\quad f'\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

$f''(x) = -\sin(x)$ $\quad f''\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

$f'''(x) = -\cos(x)$ $\quad f'''\left(\frac{3\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$f(x) \approx \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\left(x-\frac{3\pi}{4}\right) - \frac{\sqrt{2}}{2}\cdot\frac{1}{2!}\left(x-\frac{3\pi}{4}\right)^2 + \frac{\sqrt{2}}{2}\frac{1}{3!}\left(x-\frac{3\pi}{4}\right)^3$

3. Find an upper bound for the error made in approximating the value of $e^{\sqrt{3}}$ using the nonzero terms through $n = 6$ in its Maclaurin series.

degree →

$f(x) = e^x$ $\quad f^{(7)}(x) = e^x$

$\max_{0 \le x \le \sqrt{3}} \left|f^{(7)}(x)\right| \le e^{\sqrt{3}} < 3^{\sqrt{3}} < 3^{\sqrt{4}} = 9$

$|\text{Error}| = |f(\sqrt{3}) - P_6(\sqrt{3})| \le \dfrac{3^{\sqrt{3}}(\sqrt{3}-0)^7}{7!} \approx 0.0622$

Your answer may be different depending on your choice of the upper bound of $\left|f^{(7)}(x)\right|$. I used 3 as upper bound of e, but I assumed I have a calculator that does all basic arithmetic such as +, −, ×, ÷, √, any root, any exponent.

4. When approximating $\ln(0.8)$, how many nonzero terms of its Maclaurin series do you have to use in order to have an error that is less than 0.0001?

$f(x) = \ln(1+x)$ $\quad f(-0.2) = \ln(0.8)$

$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$ (for $|x|<1$)

$= \sum_{n=1}^{\infty} (-1)^{n+1}\dfrac{x^n}{n}$

$f(-0.2) = -0.2 - \frac{(0.2)^2}{2} - \frac{(0.2)^3}{3} - \ldots$

Not alternating, can't use alt. ser. error bound.

$f'(x) = \frac{1}{1+x} = (1+x)^{-1}$

$f''(x) = -1(1+x)^{-2}$

$f^{(3)}(x) = 2(1+x)^{-3}$

$f^{(n)}(x) = (n-1)!\,(1+x)^{-n}$

$f^{(n+1)}(x) = n!\,(1+x)^{-(n+1)} = \dfrac{n!}{(1+x)^{n+1}}$

$\max_{-0.2 \le x \le 1} \left|f^{(n+1)}(x)\right| \le \dfrac{n!}{(1-0.2)^{n+1}} = \dfrac{n!}{(0.8)^{n+1}}$

$\text{Error} \le \left|\dfrac{n!}{(0.8)^{n+1}} \cdot \dfrac{(-0.2-0)^{n+1}}{(n+1)!}\right| < 0.0001$

n	$\frac{(0.2-0)^{n+1}}{n+1}$
4	0.0002
5	0.000041

up to degree 5 uses 5 terms

$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

$\cos(2x) = 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!}$ $\quad f(x) = \cos(2x)$

5. What are the possible values for x if we want the error in approximating $\cos(2x)$ using the first four nonzero terms of its Maclaurin series to be less than 0.01?

degree 6

Alternating series (for any x)

$\left|\frac{(2x)^{2n}}{(2n)!}\right|$ decreases towards 0 as $n \to \infty$ for any x.

Alternating series error bound, "next term" is the 8th degree term.

$|\text{error}| < \frac{(2x)^8}{8!} \le 0.01 \Rightarrow 256x^8 \le 403.2 \Rightarrow -\sqrt[8]{\frac{403.2}{256}} \le x \le \sqrt[8]{\frac{403.2}{256}}$

$-1.058 \le x \le 1.058$

OR you can use Lagrange → Degree 6, BUT also degree 7 since the 7th degree term is zero. So we can look at $f^{(8)}(x)$

$|f^{(8)}(x)| = |2^8 \cos(2x)| \le 2^8 = 256 \Rightarrow |\text{Error}| \le \frac{256}{8!}|x|^8 < 0.01$

$-1.058 < x < 1.058$

6. When approximating $\tan^{-1}(0.75)$, how many nonzero terms of its Maclaurin series do you have to use in order to have an error that is less than 0.0001?

As $n \to \infty$, $|a_n| = \frac{0.75^{2n+1}}{2n+1}$ decreases towards 0.

$f(x) = \arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$ ← Careful!! n starts at 0 here!

Use $|x| < 1$

$f(0.75) = \arctan(0.75) = \sum_{n=0}^{\infty} \frac{(0.75)^{2n+1}}{2n+1}(-1)^n$ ← Alternating

Alternating Series Error

$|\text{Error}| = |f(0.75) - P_n(0.75)| < |a_{n+1}| \le 0.0001$

$|a_{n+1}| = \frac{0.75^{2n+3}}{2n+3}$

n	a_{n+1}
9	0.00011
10	0.000058

$a_{11} < 0.0001$

Need terms $a_0 + a_1 + \dots + a_{10}$ (11 terms)

Look at the sigma notation, notice my n is NOT the degree here!! It is (almost) the number of terms.

My n is actually the "number of terms PLUS ONE" since my summation starts at n=0.

7. Suppose $f(2) = 1$, $f'(2) = 2$, $f''(2) = 4$, $f'''(2) = 18$, and $\left|f^{(n)}(x)\right| < 6$ for all x and for $n > 3$.

↑ centered at 2

a. Write the first four nonzero terms of the Taylor series for f expanded about $x_0 = 2$.

$P_3 = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$

$= 1 + 2(x-2) + \frac{4}{2!}(x-2)^2 + \frac{18}{3!}(x-2)^3$

b. Find an upper bound for the error made in approximating the value of $f(2.3)$ using the first four nonzero terms of Taylor series for f expanded about $x_0 = 2$ from part (a).

$|f^{(4)}(x)| < 6$ (given)

$|\text{Error}| = |f(2.3) - P_3(2.3)| \le \frac{6(2.3-2)^4}{4!} = \frac{81}{4000} = 0.002025$

8. Use Maclaurin series to evaluate $\lim_{x \to 0} \frac{\ln(1+x)}{\sin(3x)}$.

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = x\left(1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots\right)$

$\sin(3x) = (3x) - \frac{(3x)^3}{3!} + \frac{(3x)^5}{5!} - \frac{(3x)^7}{7!} + \dots = 3x\left(1 - \frac{(3x)^2}{3!} + \frac{(3x)^4}{5!} - \frac{(3x)^6}{7!} + \dots\right)$

$\lim_{x \to 0} \frac{\ln(1+x)}{\sin(3x)} = \lim_{x \to 0} \frac{x\left(1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots\right)}{3x\left(1 - \frac{(3x)^2}{3!} + \frac{(3x)^4}{5!} - \frac{(3x)^6}{7!} + \dots\right)} = \frac{1}{3}$

9. Use power series to approximate the value of $\int_0^{0.9} \sin\left(x^2\right) dx$ with error less than 0.0001.

$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

$\sin(x^2) = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \frac{x^{14}}{7!} + \dots = \sum_{n=0}^{\infty} \frac{x^{2(2n+1)}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{x^{4n+2}(-1)^n}{(2n+1)!}$

According to my sigma notation here, just like #6, my n stands for "number of terms plus one", not the degree!

$\int_0^{0.9} \sin(x^2)\,dx = \left[\frac{1}{3}x^3 - \frac{1}{7}\frac{x^7}{3!} + \frac{1}{11}\frac{x^{11}}{5!} - \frac{1}{15}\frac{x^{15}}{7!} + \dots\right]_0^{0.9}$

$= \sum_{n=0}^{\infty} \left[\frac{x^{4n+3}(-1)^n}{(4n+3)(2n+1)!}\right]_0^{0.9} = \sum_{n=0}^{\infty} \frac{(0.9)^{4n+3}(-1)^n}{(4n+3)(2n+1)!}$ ← Notice n starts at 0 here!

Alternating Series | nth non zero term | $= |a_n| = \frac{(0.9)^{4n+3}}{(4n+3)(2n+1)!}$

$|\text{error}| < |a_{n+1}| = \frac{(0.9)^{4(n+1)+3}}{(4(n+1)+3)(2(n+1)+1)!} = \frac{(0.9)^{4n+7}}{(4n+7)(2n+3)!}$

n	$\lvert a_{n+1}\rvert$
1	0.00024
2	0.0000027 < 0.0001

$(n+1)=3$

$\Rightarrow$ need $n=2$, but our n starts at 0

So, need 3 terms $(a_0 + a_1 + a_2)$

$\Rightarrow \int_0^{0.9} \sin(x^2)\,dx \approx \frac{1}{3}(0.9)^3 - \frac{1}{7}\frac{(0.9)^7}{3!} + \frac{1}{11}\frac{(0.9)^{11}}{5!}$

≈ 0.23185

10. Suppose that f is a function such that $f(1)=1$, $f'(1)=2$, and $f''(x) = \frac{1}{(1+x^3)}$ for all $x > -1$.

Centered at 1

a. Estimate the value of $f(1.5)$ using a quadratic Taylor polynomial. $f''(1) = \frac{1}{2}$

$P_2 = 1 + 2(x-1) + \frac{1/2}{2!}(x-1)^2 = 1 + 2(x-1) + \frac{1}{4}(x-1)^2$

$f(1.5) \approx P_2(1.5) = 1 + 2(0.5) + \frac{1}{4}(0.5)^2 = 1 + 1 + \frac{1}{16} = \frac{33}{16}$

b. Determine an upper bound for the error made in your approximating in part (a).

$f^{(3)}(x) = -(1+x^3)^{-2}(3x^2) = \frac{-3x^2}{(1+x^3)^2}$

chain rule!

$\frac{d}{dx}\left[f^{(3)}(x)\right] = \frac{(1+x^3)^2(-6x) - (-3x^2)2(1+x^3)^1 3x^2}{(1+x^3)^4}$

$\text{Top} = -6x(1+x^3)^2 + 18x^4(1+x^3)$

$= -6x(1+x^3)[1+x^3-3x^3]$

$= -6x(1+x^3)(1-2x^3) > 0$ for $1 \le x \le 1.5$

$f^{(3)}(x) < 0$ and is increasing for $1 \le x \le 1.5$

$\max_{1\le x\le 1.5} \left|f^{(3)}(x)\right| \le \left|f^{(3)}(1)\right| = \frac{3}{4}$

$|\text{error}| \le \frac{\frac{3}{4}(1.5-1)^3}{3!} = \frac{1}{64}$

11. We can show that the series $\sum_{n=0}^{\infty} \frac{n^2}{2^n}$ converges by the Ratio Test, but to what value does it converge? To find out, complete the following steps.

Express $\frac{1}{1-x}$ as a series: $\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \ldots$

$(1-x)^{-1}$

Differentiate both sides.

$(1-x)^{-2}(-1) = 1 + 2x + 3x^2 + 4x^3 + \ldots$

$\frac{1}{(1-x)^2}$

Multiply both sides by x.

$\frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + 4x^4 + \ldots$

Differentiate both sides (again).

$\frac{(1-x)^2(1) - (x)2(1-x)(-1)}{(1-x)^4} = 1 + 4x + 9x^2 + 16x^3 + \ldots$

Simplify $\frac{(1-x)[1-x+2x]}{(1-x)^{4-3}} = \frac{1+x}{(1-x)^3}$

Multiply both sides by x (again) and express the right-hand side in simplified Σ-notation

$\frac{x+x^2}{(1-x)^3} = x + 4x^2 + 9x^3 + 16x^4 + \ldots = \sum_{n=1}^{\infty} n^2 x^n$

Substitute $\frac{1}{2}$ for x and simplify both sides of your equation. What result do you get?

$\frac{\frac{1}{2} + \left(\frac{1}{2}\right)^2}{\left(1-\frac{1}{2}\right)^3} = \frac{\frac{3}{4}}{\frac{1}{8}} = 6 = \sum_{n=1}^{\infty} n^2\left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \frac{n^2}{2^n}$